Gaussian Parcel Models for Fluid Particle Interaction

Physical formulation coupling forces collisions and implementation

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1 Identification and scope

The most defensible general name for the physics described in the request is **an Eulerian–Lagrangian parcel model with Gaussian regularization of interphase coupling**, supplemented by a specified particle collision model. The supplied picture establishes an incompressible carrier fluid, tracked particle or parcel positions, drag, and a normalized feedback kernel. It does **not** specify that the kernel is Gaussian, or define a collision law. Consequently, the picture alone cannot establish one unique named method.

Several established methods match different parts of the description:

Interpretation of the description	Relevant established framework
A parcel represents many particles; its influence on the fluid is spread spatially	Eulerian–Lagrangian parcel tracking with Gaussian interpolation or volume filtering
A physical particle is represented by Gaussian force envelopes and force multipoles	Force Coupling Method, FCM
Gaussian influence regions determine statistical collisions between parcels	Gaussian or ellipsoidal parcel collision algorithms, including Pischke and coauthors
Dense parcels interact through a continuum particle stress and statistical velocity relaxation	Multiphase Particle in Cell, MP-PIC
A small point particle produces a regularized unsteady viscous disturbance	Exact Regularized Point Particle method, ERPP

The closest single fluid–parcel–collision paper found is Apte, Mahesh, and Lundgren (2008), which combines Gaussian grid transfer, parcel dynamics, fluid displacement effects, and a separate interparcel collision model [R1]. If the Gaussian specifically describes the collision influence region, Pischke, Cordes, and Kneer (2012) is the more targeted reference [R3]. If the remembered term was a Gaussian force envelope calibrated to a sphere, investigate FCM [R4–R6]. These are candidate identifications, not proof of which paper or method the user remembers.

This note provides a self-contained physical specification independent of any neural-network solver. Sections 2–10 construct a baseline continuum/parcel model. Sections 11–15 specify alternative collision and interaction closures. Later sections explain FCM, MP-PIC, numerical solution, and validation. Equations identified as constructions or derivations are the formulation developed here; they are not presented as verbatim implementations of a cited paper. No numerical simulation has been performed for this note.

2 Physical meaning of a parcel

Let parcel i represent N_i spherical physical particles with diameter d_i, material density $\rho_{p,i}$, and representative velocity $\mathbf{v}_i$. Define

$$V_{p,i} = \frac{\pi d_i^3}{6}, \qquad m_{p,i} = \rho_{p,i} V_{p,i}, \qquad M_i = N_i m_{p,i}.$$

The distinction between $m_{p,i}$ and M_i is essential. Drag correlations use the diameter of a **physical particle**. The total force transferred by the parcel is multiplied by its multiplicity N_i.

For identical deterministic forcing on its members,

$$M_i \dot{\mathbf{v}}_i = N_i \mathbf{F}_{p,i} \quad \Longrightarrow \quad \dot{\mathbf{v}}_i = \frac{\mathbf{F}_{p,i}}{m_{p,i}}.$$

Increasing the number represented does not, by itself, increase the relaxation time of the physical particles. A statistical parcel is not automatically a rigid aggregate, a large solid sphere, or a permanently bound cluster. Its members are assumed to be sufficiently similar for the quantities being predicted. Different sizes, crossing

velocity populations, or strong internal dispersion can require parcel splitting or additional distribution variables. Multiplicity can be a statistical weight rather than a literal integer in weighted numerical methods.

3 What Gaussian means

3.1 A normalized spatial kernel

For D spatial dimensions, define a smoothing width h_i and

$$G_i(\mathbf{x}) = \frac{1}{(2\pi h_i^2)^{D/2}} \exp\left[-\frac{|\mathbf{x}-\mathbf{q}_i|^2}{2h_i^2}\right], \qquad \int_{\mathbb{R}^D} G_i \, d\mathbf{x} = 1.$$

Here $\mathbf{q}_i$ is the parcel center. In three dimensions G_i has units m^{-3}. Its purpose may be to smooth a force, estimate a concentration, or approximate a cloud density. These interpretations are different modeling choices. A numerical smoothing width does not automatically equal a particle radius or the physical spread of the parcel.

In a bounded domain a simple normalized construction is

$$W_i(\mathbf{x}) = \frac{G_i(\mathbf{x})}{Z_i}, \qquad Z_i = \int_\Omega G_i(\mathbf{x}) \, d\mathbf{x}.$$

Then $\int_\Omega W_i \, d\mathbf{x} = 1$. This conserves the total deposited quantity. It does not by itself reproduce wall hydrodynamics or preserve the kernel's first moment near a boundary. Periodic domains require periodic images, or an equivalently periodized kernel. Practical Gaussian stencils are truncated; they must be renormalized after truncation.

3.2 An anisotropic cloud

For a symmetric positive definite covariance $\mathbf{H}_i$,

$$G_i(\mathbf{x}) = \frac{\exp[-\frac{1}{2}(\mathbf{x}-\mathbf{q}_i)^T \mathbf{H}_i^{-1}(\mathbf{x}-\mathbf{q}_i)]}{(2\pi)^{D/2}\sqrt{\det \mathbf{H}_i}}.$$

Its principal axes describe an ellipsoidal influence region. Anisotropic parcel collision methods use such geometry to improve density estimation in sheets, jets, or other nonuniform populations [R3]. This is separate from assuming that particle **velocities** are Gaussian.

3.3 Four different uses of a Gaussian

1. **Spatial regularization:** distribute a point force over nearby fluid locations.
2. **Spatial population density:** approximate the physical distribution of particles inside a cloud.
3. **Velocity distribution:** approximate unresolved particle velocity fluctuations, for example in a kinetic collision model.
4. **Random forcing:** describe Brownian or modeled turbulent fluctuations.

A spatial Gaussian does not imply the other three. In particular, two smooth Gaussian envelopes may overlap without any physical solid particles touching.

4 State variables and assumptions

The baseline uses a Newtonian, constant-density carrier fluid and inertial, rigid, heavy spherical particles. Particle volume fraction is small enough that fluid displacement can initially be neglected. Mass loading and two-way momentum feedback can still be important because ρ_p/ρ_f can be large.

Quantity	Definition	Three dimensional SI units
$\mathbf{u}, p$	Fluid velocity and pressure	m/s, Pa
ρ_f, μ, ν	Fluid density, dynamic and kinematic viscosity	kg/m^3, Pa s, m^2/s
$\mathbf{q}_i, \mathbf{v}_i$	Parcel center and representative velocity	m, m/s
N_i, M_i	Multiplicity and represented mass	dimensionless, kg
$\mathbf{F}_{h,i}$	Total hydrodynamic force on parcel i	N
$\mathbf{f}_{p\to f}$	Force on fluid per mixture volume	N/m^3
τ_i, h_i	Particle response time and kernel width	s, m

The initial baseline excludes thermal effects, phase change, adhesion, particle rotation, Brownian motion, and unresolved turbulence. These are extensions, not effects generated automatically by Gaussian smoothing. Collision closure must be selected explicitly. In a two dimensional model, specify whether masses and forces are per unit depth; do not insert a three dimensional mass into a two dimensional force density without a thickness convention.

5 Carrier fluid equations

For the dilute baseline,

$$\nabla \cdot \mathbf{u} = 0,$$

$$\rho_f \left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} \right] = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}_{p\to f} + \mathbf{f}_{\text{ext}}.$$

The stress form is

$$\sigma_f = -p\mathbf{I} + \mu[\nabla \mathbf{u} + (\nabla \mathbf{u})^T], \qquad \rho_f D_t \mathbf{u} = \nabla \cdot \sigma_f + \mathbf{f}_{p\to f} + \mathbf{f}_{\text{ext}}.$$

External forcing may include a specified pressure drive or fluid gravity, with its hydrostatic pressure handled consistently. These equations can be solved by finite volumes, finite elements, or a spectral method. The physical model does not prescribe a neural-network architecture.

6 Parcel trajectories and hydrodynamic forces

Parcel translation satisfies

$$\dot{\mathbf{q}}_i = \mathbf{v}_i, \qquad M_i \dot{\mathbf{v}}_i = \mathbf{F}_{h,i} + \mathbf{F}_{g,i} + \mathbf{F}_{c,i} + \mathbf{F}_{w,i} + \mathbf{F}_{\text{other},i}.$$

$\mathbf{F}_{c,i}$ and $\mathbf{F}_{w,i}$ are particle and wall contact forces when a continuous-force collision model is used. With instantaneous hard-sphere collisions, use velocity jumps instead of a continuous $\mathbf{F}_c$.

6.1 Drag and response time

Define a sampled fluid velocity and slip,

$$\mathbf{U}_i = \mathcal{J}_i \mathbf{u} = \int_\Omega W_i \mathbf{u}\, d\mathbf{x}, \qquad \mathbf{w}_i = \mathbf{U}_i - \mathbf{v}_i.$$

For the baseline regularized drag model,

$$\mathbf{F}_{D,i} = \frac{M_i}{\tau_i}\mathbf{w}_i, \qquad \tau_i = \frac{\rho_{p,i} d_i^2}{18\mu}.$$

This response time is the Stokes result for small slip Reynolds number. More generally, a physical-particle drag correlation has the form

$$\mathbf{F}_{D,i} = N_i \frac{1}{2}\rho_f C_D A_{p,i} |\mathbf{w}_i| \mathbf{w}_i, \quad A_{p,i} = \frac{\pi d_i^2}{4}, \quad Re_{p,i} = \frac{\rho_f d_i |\mathbf{w}_i|}{\mu}.$$

For example, the commonly used Schiller–Naumann form is

$$C_D = \frac{24}{Re_p}(1 + 0.15 Re_p^{0.687})$$

over its usual moderate-Re_p range, approximately below 10^3, for isolated spheres [R15]. Its $Re_p \to 0$ limit should be evaluated through the finite Stokes drag expression, not by dividing numerically by zero. Concentrated suspensions require a validated voidage-dependent law; a dilute isolated-sphere correlation is not universal. Official coupled CFD–DEM documentation gives examples of Gaussian distribution and alternative hydrodynamic closures [R12].

6.2 Disturbed and undisturbed fluid velocity

The resolved field already contains the response to the particle's own deposited force. Classical isolated-particle drag is based on the **undisturbed** fluid velocity, meaning the field with that particle's own disturbance removed. A physically calibrated point-particle implementation may therefore use

$$\mathbf{U}_i^{\infty} = \mathcal{J}_i \mathbf{u} - \mathbf{U}_i^{\mathrm{self}}.$$

ERPP and drag-correction studies address this issue explicitly [R7,R8]. The baseline $\mathbf{U}_i = \mathcal{J}_i \mathbf{u}$ is a fully defined regularized drag coupling, useful for conservative implementation and verification. It should not be claimed to reproduce an isolated real sphere accurately for arbitrary h_i/d_i. Calibrate its mobility or adopt a published self-disturbance correction when that accuracy matters. An altered slip definition also changes the energy argument in Section 8.

6.3 Gravity and buoyancy

For the heavy-particle baseline, use $\mathbf{F}_{g,i} = M_i \mathbf{g}$. When the displaced-fluid weight is important and the hydrostatic contribution has been removed from other particle-force terms, use

$$\mathbf{F}_{g+b,i} = N_i(\rho_{p,i} - \rho_f) V_{p,i} \mathbf{g}.$$

Alternatively, retain the full fluid pressure force and $M_i \mathbf{g}$. Do not then add buoyancy a second time. Near-neutral particles require a more complete fluid-force and fluid-displacement model.

7 Gaussian feedback from parcels to fluid

With $\mathbf{F}_{h,i}$ defined positive **on the parcel**, the force on the fluid is

$$\boxed{\mathbf{f}_{p\to f}(\mathbf{x}, t) = -\sum_i \mathbf{F}_{h,i}(t) W_i(\mathbf{x}, t).}$$

For drag-only coupling,

$$\mathbf{f}_{p\to f} = -\sum_i \frac{M_i}{\tau_i}(\mathbf{U}_i - \mathbf{v}_i)W_i.$$

Normalization gives the exact integral identity

$$\int_\Omega \mathbf{f}_{p\to f}\, d\mathbf{x} = -\sum_i \mathbf{F}_{h,i}.$$

Thus every momentum increment delivered by drag to the particles has an equal and opposite fluid increment. A pairwise particle contact transfers momentum between particles. It is not itself a hydrodynamic force and should not be deposited again as fluid drag. Gravity is an external force, not a particle-to-fluid reaction. Wall impacts transfer momentum to the wall unless wall motion is also solved.

This force-spreading structure is the part of the supplied image that can be identified confidently. Gaussian spreading and the collision law are separate decisions. Apte and coauthors give one published combination; other combinations are possible [R1].

8 Conservative discrete transfer and energy

For cell a, volume ΔV_a, and center $\mathbf{x}_a$, define

$$D_{ai} = \frac{G_i(\mathbf{x}_a)}{\sum_b G_i(\mathbf{x}_b)\Delta V_b}, \qquad \sum_a D_{ai}\Delta V_a = 1.$$

Use the same weights for interpolation and spreading:

$$\mathbf{U}_i = \sum_a D_{ai}\mathbf{u}_a \Delta V_a, \qquad \mathbf{f}_a = -\sum_i D_{ai}\mathbf{F}_{h,i}.$$

This gives both discrete force conservation and a work identity,

$$\sum_a \mathbf{u}_a \cdot \mathbf{f}_a \Delta V_a = -\sum_i \mathbf{U}_i \cdot \mathbf{F}_{h,i}.$$

For the baseline Stokes coupling, the combined instantaneous power delivered to fluid and parcels by drag is therefore

$$P_D = \sum_i (\mathbf{v}_i - \mathbf{U}_i)\cdot \mathbf{F}_{D,i} = -\sum_i \frac{M_i}{\tau_i}|\mathbf{U}_i - \mathbf{v}_i|^2 \le 0.$$

This is a derivation for the specified model: drag dissipates relative kinetic energy. It is a semidiscrete identity, not a guarantee for an arbitrary time integrator. Use identical impulse quadrature in both phases. Adjoint transfer does not automatically guarantee angular momentum conservation, first-moment accuracy, correct wall mobility, or correct self-disturbance subtraction.

9 Concentration and finite particle volume

From the parcel representation one can construct

$$n(\mathbf{x}) = \sum_i N_i W_i, \qquad \rho_s^{\mathrm{bulk}}(\mathbf{x}) = \sum_i M_i W_i, \qquad \alpha_s(\mathbf{x}) \approx \sum_i N_i V_{p,i} W_i.$$

Here n is number density, and ρ_s^{bulk} is solid mass per mixture volume, not material density. In the unbounded isotropic fixed-width model,

$$\mathbf{j}_s = \sum_i N_i V_{p,i} \mathbf{v}_i W_i, \qquad \partial_t \alpha_s + \nabla \cdot \mathbf{j}_s = 0.$$

The identity follows directly from $\partial_t W_i = -\mathbf{v}_i \cdot \nabla W_i$. Moving widths, truncated boundary kernels, and approximate deposition require additional care. A large value of $\sum_i N_i V_{p,i} W_i$ can exceed one: Gaussian smoothing is not a packing constraint. Arbitrary clipping destroys represented volume unless accompanied by a conservative redistribution model.

If particle displacement is significant, set $\alpha_f = 1 - \alpha_s$ and replace the dilute fluid continuity equation by

$$\partial_t(\alpha_f \rho_f) + \nabla \cdot (\alpha_f \rho_f \mathbf{u}_f) = 0.$$

For constant ρ_f, this is $\partial_t \alpha_f + \nabla \cdot (\alpha_f \mathbf{u}_f) = 0$. It is generally not equivalent to $\nabla \cdot \mathbf{u}_f = 0$ when α_f varies. A volume-filtered momentum equation also requires consistent pressure partition, interphase forces, and unresolved stresses. Capecelatro and Desjardins derive these terms and discuss their closures [R2]. Adding only a concentration field to the dilute Navier–Stokes solver is not a complete dense-flow model.

One reduced pressure-partition convention uses a fluid pressure term $-\alpha_f \nabla p$ and a per-physical-particle ambient pressure force $-V_p \nabla p$, alongside drag. In that convention the fluid receives drag reaction, while the two phase pressure contributions together give approximately $-\nabla p$. Depositing the same ambient pressure force again on the fluid double-counts that contribution. Follow one complete published volume-averaged convention throughout; do not mix its force source with the total-force convention in Section 7.

10 Hydrodynamic interaction and contact are different

Parcel i changes the fluid through its feedback. Parcel j then samples that changed fluid and its drag changes. This is fluid-mediated interaction. It requires no direct mechanical contact, and a wake can influence another parcel well beyond a Gaussian force stencil because the fluid equations transmit the disturbance.

Direct contact requires a separate physical law or statistical collision operator. One-way coupling means fluid affects particles only. Two-way coupling includes particle feedback. The commonly used four-way classification additionally includes particle–particle interactions or collisions. Its applicability depends on both loading and collision timescales, not simply the number of tracked parcels.

For low-Reynolds-number liquid suspensions, close approaches can also require a lubrication correction before solid contact occurs. A Gaussian force distribution does not resolve arbitrarily thin fluid gaps. Lubrication corrections are explicitly developed in the FCM literature [R6].

11 Collision option A physical hard spheres

This section is a complete frictionless binary collision rule for individual physical spheres, $N_i = N_j = 1$. Let

$$\mathbf{n} = \frac{\mathbf{q}_i - \mathbf{q}_j}{|\mathbf{q}_i - \mathbf{q}_j|}, \quad g_n = (\mathbf{v}_i - \mathbf{v}_j) \cdot \mathbf{n}, \quad r_i = d_i/2.$$

Contact occurs at $|\mathbf{q}_i - \mathbf{q}_j| = r_i + r_j$. Apply an impact only if $g_n < 0$, meaning the pair is approaching. For restitution $0 \le e_n \le 1$,

$$J = -\frac{(1+e_n)g_n}{1/m_{p,i} + 1/m_{p,j}},$$

$$\mathbf{v}_i^+ = \mathbf{v}_i^- + \frac{J}{m_{p,i}}\mathbf{n}, \qquad \mathbf{v}_j^+ = \mathbf{v}_j^- - \frac{J}{m_{p,j}}\mathbf{n}.$$

The pair momentum is conserved. The change in kinetic energy is

$$\Delta E = -\tfrac{1}{2}\mu_{ij}(1-e_n^2)g_n^2, \qquad \mu_{ij} = \frac{m_{p,i}m_{p,j}}{m_{p,i}+m_{p,j}}.$$

For stationary frictionless walls, reverse and scale the incoming normal component by e_w; leave tangential velocity unchanged. Use moving-wall relative velocity if appropriate. Detect impact time or use sufficiently small substeps to avoid tunneling and repeated impacts during an unresolved overlap.

Do not apply this geometric rule to centers of statistical parcels and claim that it represents all physical collisions. Their unresolved members occupy an unknown distribution. Treating each entire parcel as one massive colliding sphere defines a different coarse-grained model and needs a calibrated contact radius, inertia, stiffness, and collision-rate scaling.

12 Collision option B statistical Gaussian parcels

This is an explicitly stated kinetic construction, not a reproduction of the full Pischke algorithm. Assume each parcel has a number-density contribution $n_i = N_i G_i$ in an unbounded domain, positions are statistically independent at the collision scale, and each parcel has a single representative velocity. For distinct parcels, define the physical hard-sphere collision kernel

$$K_{ij} = E_{ij}\pi(r_i + r_j)^2|\mathbf{v}_i - \mathbf{v}_j|,$$

where E_{ij} is a specified collision efficiency. K_{ij} has units volume per time. For isotropic Gaussian spatial distributions their overlap is

$$I_{ij} = \int G_i G_j \, d\mathbf{x} = \frac{\exp[-|\mathbf{q}_i - \mathbf{q}_j|^2/(2s_{ij}^2)]}{(2\pi s_{ij}^2)^{3/2}}, \quad s_{ij}^2 = h_i^2 + h_j^2.$$

Consequently, under these assumptions the expected number of physical collisions during a short interval is

$$\Lambda_{ij} \approx N_i N_j K_{ij} I_{ij} \Delta t.$$

This follows by integrating $K_{ij} n_i n_j$ over space and time. Λ is a count, not a probability, and may exceed one. Sum unordered pairs only once. For normalized bounded kernels evaluate $I_{ij} = \int_\Omega W_i W_j d\mathbf{x}$; the unbounded closed form generally no longer applies. With anisotropic unbounded Gaussians, replace $s_{ij}^2\mathbf{I}$ by $\mathbf{H}_i + \mathbf{H}_j$.

12.1 A restricted equal multiplicity sampling rule

For $N_i = N_j = N$, a representative collision event can stand for N physical binary impacts. Its rate is therefore

$$\lambda_{ij} = N K_{ij} I_{ij}, \qquad p_{ij} \approx \lambda_{ij}\Delta t \ll 1.$$

On acceptance, sample a collision normal from the hard-sphere impact distribution, then apply the physical-mass impulse rule of Section 11 to both representative velocities. Multiplying both physical momenta by the common N preserves represented momentum exactly. The event rate times N gives $N^2 K_{ij} I_{ij}$ physical impacts in expectation.

One normal-sampling construction is: let $\mathbf{e} = (\mathbf{v}_i - \mathbf{v}_j)/|\mathbf{v}_i - \mathbf{v}_j|$; sample an azimuth uniformly and a transverse impact vector $\mathbf{b}$ uniformly over the unit disk perpendicular to $\mathbf{e}$. Then choose

$$\mathbf{n} = -\sqrt{1 - |\mathbf{b}|^2}\,\mathbf{e} + \mathbf{b}.$$

This ensures an approaching impact. The parcel-center separation is a density-estimation distance here, not the unresolved physical impact vector. If relative speed is zero, the stated monokinetic kernel gives no collisions.

Subcycle and recompute rates after impacts. $1 - e^{-\lambda \Delta t}$ gives the probability of at least one event at a frozen rate, but a single velocity update does not represent multiple impacts accurately when $\lambda \Delta t$ is large. The equal-weight construction is only a baseline statistical closure; it needs collision-rate and distribution validation.

12.2 What a production parcel collision algorithm still needs

Unequal multiplicities require conservative splitting, redistribution, or a validated weighted collision rule. A collision normal cannot simply be taken from parcel centers in a stochastic density model. Same-parcel collisions and nonzero within-parcel velocity spread are absent from the monokinetic construction. Introduce velocity distributions or more parcels if these matter. Dense spatial correlations, correlated turbulence, strong mean gradients, coalescence, and fragmentation require additional modeling.

The Pischke literature addresses Gaussian/ellipsoidal influence regions and numerical collision-estimation errors [R3]. The super-droplet method supplies a separate established treatment of weighted stochastic coalescence for droplets [R11]. Its multiplicity handling is relevant, but droplet coalescence is not the same physics as elastic solid impacts.

13 Collision option C soft sphere contact and rotation

For actual spheres or explicitly defined coarse-grained contact objects, introduce

$$\delta_{ij} = r_i + r_j - |\mathbf{q}_i - \mathbf{q}_j|, \qquad g_n = (\mathbf{v}_i - \mathbf{v}_j) \cdot \mathbf{n}.$$

A simple nonadhesive normal force on i, active only for $\delta_{ij} > 0$, is

$$\mathbf{F}_{n,ij} = \max(0, k_n \delta_{ij} - \gamma_n g_n)\mathbf{n}, \qquad \mathbf{F}_{n,ji} = -\mathbf{F}_{n,ij}.$$

The damping term is repulsive during approach. Clipping avoids an unphysical attractive tail. Hertzian contact instead uses an elastic term proportional to $\delta^{3/2}$ with a compatible damping model. Normal restitution depends on the complete chosen contact law and its integration; it cannot be assigned independently of damping without calibration.

Tangential friction needs the relative velocity at the contact points, including spin, and usually a transported tangential displacement history ξ_t. A standard structure is a trial spring–dashpot force

$$\mathbf{F}_t^{\text{trial}} = -k_t \xi_t - \gamma_t \mathbf{v}_t, \qquad |\mathbf{F}_t| \le \mu_c |\mathbf{F}_n|.$$

Rotate the history into the current contact plane and adjust it when sliding occurs. The contact force produces torque through its lever arm. For a sphere,

$$I_p = \tfrac{2}{5} m_p r^2, \qquad I_p \dot{\omega} = \mathbf{T}_h + \sum \mathbf{r}_c \times \mathbf{F}_t.$$

For independent identical members, represented inertia is $N_i I_{p,i}$; a statistical cloud is not a rigid object with moment of inertia determined by its Gaussian width. LAMMPS documents distinct normal, tangential, rolling, and twisting contact laws and their parameter conventions [R10]. A complete implementation must select one law and resolve its contact time, often much shorter than the fluid timestep. Contact stiffness has units N/m for the linear law; damping has units kg/s in the convention above.

14 A smooth Gaussian repulsion is a separate model

If the intended model treats each parcel as a smooth interacting blob, a conservative surrogate can be constructed from the overlap:

$$\mathcal{U}_{ij} = \kappa_{ij} I_{ij}, \qquad \mathbf{F}_{ij} = -\nabla_{\mathbf{q}_i} \mathcal{U}_{ij} = \frac{\kappa_{ij} I_{ij}}{s_{ij}^2} (\mathbf{q}_i - \mathbf{q}_j),$$

with $\kappa_{ij} = \kappa_{ji} > 0$ and $\mathbf{F}_{ji} = -\mathbf{F}_{ij}$. In three dimensions κ has units $\mathrm{J\,m^3}$, so $\mathcal{U}$ is energy and $\mathbf{F}$ is force. The force pushes centers apart. This formula is an original potential-based construction for fixed isotropic widths, not a standard consequence of parcel tracking and not a hard-sphere collision law.

It gives a finite energy barrier, has nonzero range at every separation, and vanishes at exact coincidence by symmetry. Sufficiently energetic blobs can interpenetrate. A damping force needs its own activation range and energy analysis. Calibrate this surrogate if used for coarse repulsion; do not label it a resolved physical impact or infer restitution, friction, or particle collision rates from it. Width-dependent potentials also require extra terms if width is a dynamic degree of freedom.

15 Internal cloud evolution and additional physics

If $\mathbf{H}_i$ is only a numerical smoothing parameter, no physical evolution equation is required; prescribe it and study sensitivity. If it describes a physical cloud, its evolution requires a closure. For a passive cloud in an affine flow with velocity gradient $\mathbf{A}$ and diffusion tensor $\mathbf{D}$, the Gaussian advection–diffusion construction gives

$$\dot{\mathbf{H}} = \mathbf{A}\mathbf{H} + \mathbf{H}\mathbf{A}^T + 2\mathbf{D}.$$

This does not automatically hold for inertial particles with collisions. An inertial cloud generally needs a phase-space covariance involving position, velocity, and their cross-correlation. Strongly nonlinear flow, trajectory crossing, or impacts can make a single Gaussian inadequate.

Additional hydrodynamic forces should be selected by regime. For a small sphere at low slip Reynolds number, the Maxey–Riley framework distinguishes ambient-fluid acceleration, added mass, viscous history, and finite-size corrections [R13]. Approximate examples per physical sphere are

$$\mathbf{F}_{AM} = \tfrac{1}{2}\rho_f V_p \left(\frac{D\mathbf{u}^\infty}{Dt} - \dot{\mathbf{v}} \right), \qquad \mathbf{T}_D = 8\pi\mu r^3 \left(\tfrac{1}{2}\nabla \times \mathbf{u}^\infty - \omega \right).$$

The first is the spherical added-mass term in the stated small-particle approximation; the material derivative follows the undisturbed fluid, not the particle trajectory. Lift, lubrication, history, and finite-size corrections have their own validity ranges. Adding them selectively does not make a high-Reynolds-number or near-wall calculation exact.

For Brownian motion of an isolated physical sphere, a fluctuation–dissipation-consistent inertial Langevin force can be written

$$m_p \, d\mathbf{v} = \text{deterministic force}\, dt + \sqrt{2 k_B T \zeta}\, d\mathbf{B}_t, \qquad \zeta = 6\pi\mu r.$$

Independent member noise sums with variance proportional to N_i, so its total-force amplitude scales as $\sqrt{N_i}$, not N_i. Correlated hydrodynamic Brownian motion needs correlated noise and, in overdamped formulations, the appropriate drift. Fluctuating FCM addresses this problem [R5]. Do not add arbitrary Gaussian noise and call it Brownian physics. Turbulent dispersion is a different closure.

16 The Force Coupling Method

FCM is a particularly relevant alternative if the remembered Gaussian describes a particle's **hydrodynamic envelope**, rather than a statistical population. In a common Stokes formulation,

$$-\nabla p + \mu\nabla^2\mathbf{u} + \sum_i \left[\mathbf{F}_i\Delta_i - \tfrac{1}{2}\mathbf{T}_i \times \nabla\Theta_i + \mathbf{S}_i \cdot \nabla\Theta_i\right] = 0, \qquad \nabla\cdot\mathbf{u} = 0.$$

Here $\mathbf{F}_i$ and $\mathbf{T}_i$ are the force and torque transmitted to the fluid, and $\mathbf{S}_i$ is a stresslet with the sign convention of this equation. Both envelopes are normalized Gaussians. For a sphere of radius r, a standard calibration is

$$\sigma_\Delta = \frac{r}{\sqrt{\pi}}, \qquad \sigma_\Theta = \frac{r}{(6\sqrt{\pi})^{1/3}}.$$

Translation, rotation, and rigidity are obtained through weighted fluid averages:

$$\mathbf{V}_i = \int \mathbf{u}\Delta_i\, dV, \qquad \Omega_i = \tfrac{1}{2}\int (\nabla\times\mathbf{u})\Theta_i\, dV,$$

$$\mathbf{E}_i = \tfrac{1}{2}\int [\nabla\mathbf{u} + (\nabla\mathbf{u})^T]\Theta_i\, dV = 0.$$

The stresslets enforce the averaged zero-strain constraint. This structure follows the standard FCM presentation in [R5]; [R4] is foundational. Unlike baseline inertial drag tracking, overdamped FCM obtains particle velocities from a hydrodynamic mobility solve. Each envelope normally represents a physical particle; it is not automatically a parcel of many particles. Near-contact lubrication and mechanical contact need additional treatment [R6]. Do not combine both formulations' drag responses without deriving the resulting mobility, or hydrodynamic resistance can be counted twice.

17 MP PIC and Gaussian velocity relaxation

For dense parcel flows, MP-PIC uses a phase-space distribution f and particle stress rather than resolving every mechanical contact. A representative kinetic structure is

$$\partial_t f + \mathbf{v}\cdot\nabla_x f + \nabla_v\cdot(\mathbf{a}f) = \mathcal{C}[f].$$

Its acceleration includes drag, gravity, pressure effects, and a packing-stress contribution such as $-\nabla p_s/(\alpha_s\rho_p)$ in a monodisperse isotropic closure. The precise pressure and source conventions must follow the chosen implementation [R9].

A Gaussian here may instead be a local equilibrium **velocity** distribution,

$$f_{eq} = \frac{n}{(2\pi\Theta)^{3/2}}\exp\left[-\frac{|\mathbf{v}-\overline{\mathbf{v}}|^2}{2\Theta}\right], \quad \Theta = \tfrac{1}{3}\langle|\mathbf{v}-\overline{\mathbf{v}}|^2\rangle.$$

A return-to-isotropy operator has the BGK form $\mathcal{C} = (f_{eq} - f)/\tau_c$. With matching moments it redistributes velocity fluctuations; inelastic cooling requires additional modeling. O'Rourke and Snider introduce such a return-to-isotropy extension to MP-PIC [R14]. This Gaussian has units of velocity variance inside its exponent, not spatial length variance. MP-PIC does not require Gaussian spatial interpolation, and a Gaussian spatial kernel alone does not define MP-PIC.

18 Relation to the supplied equations

Let $L, U, L/U$ define the length, velocity, and time scales. Define

$$Re = \frac{\rho_f U L}{\mu}, \qquad St_i = \frac{\tau_i U}{L}, \qquad m_i^* = \frac{M_i}{\rho_f L^3}, \qquad W_i' = L^3 W_i.$$

The dilute dimensionless drag model becomes

$$\nabla' \cdot \mathbf{u}' = 0, \qquad \partial_{t'} \mathbf{u}' + (\mathbf{u}' \cdot \nabla')\mathbf{u}' = -\nabla' p' + Re^{-1} \nabla'^2 \mathbf{u}' + \mathbf{S}',$$

$$\dot{\mathbf{q}}_i' = \mathbf{v}_i', \qquad \dot{\mathbf{v}}_i' = \frac{\mathbf{U}_i' - \mathbf{v}_i'}{St_i} + \mathbf{a}_{\text{other},i}',$$

$$\boxed{\mathbf{S}' = -\sum_i m_i^* \frac{\mathbf{U}_i' - \mathbf{v}_i'}{St_i} W_i'.}$$

This is the feedback structure shown in the picture, allowing either point evaluation or a specified averaging operator for the local fluid velocity. If $\beta = \sum_i M_i/(\rho_f|\Omega|)$ is total mass loading and four represented masses are equal, then $m_i^* = \beta|\Omega'|/4$. That equality needs this definition of β and equal masses; it is not a general parcel law.

No particle collision law follows from the drag equation. No Gaussian width follows from normalization alone. These are precisely the missing physical specifications that must be added to the picture.

19 A complete baseline solution procedure

For a concrete first implementation choose a three dimensional periodic domain, small solid volume fraction, heavy spheres, constant ρ_f and μ, fixed normalized Gaussian widths, equal multiplicity, and the statistical collision closure of Section 12. Exclude gravity for the initial momentum test. This is a closed approximate model under its stated dilute and monokinetic assumptions.

1. **Specify inputs.** Domain sizes, mesh, fluid viscosity and density, particle diameters and densities, common multiplicity, initial centers and velocities, widths, restitution, collision efficiency, final time, tolerances, and random seed. Set an initial divergence-free fluid field and fix the pressure gauge.
2. **Build periodic stencils.** Periodize/truncate the Gaussian, then normalize it with actual cell volumes. Verify the discrete unit-integral condition separately for every parcel.
3. **Interpolate fluid velocity.** Evaluate $\mathbf{U}_i$ with the same weights that will spread the feedback. If using a physically calibrated self-disturbance correction, implement and validate that model explicitly.
4. **Advance drag and fluid consistently.** Solve parcel drag and the fluid momentum equation with equal and opposite integrated impulses. A coupled implicit or converged iterative treatment is preferable for stiff drag or high mass loading. Perform the incompressibility pressure solve with boundary conditions consistent with the periodic domain.
5. **Move centers.** Integrate $\dot{\mathbf{q}}_i = \mathbf{v}_i$ using the same temporal accuracy as the selected coupling scheme; wrap periodic positions and rebuild weights when required.
6. **Sample collisions in substeps.** Compute periodized overlap integrals and pair rates. Process unordered candidate pairs in a randomized order; use short substeps so accepted events are rare for each particle population. Sample the physical collision normal and update both representative velocities conservatively. Recompute rates as states change. Avoid a cutoff that silently loses a material part of the Gaussian overlap.
7. **Reconcile coupling.** For strong coupling, repeat the fluid/parcel update to convergence or use a monolithic step. A single lagged pass is only a first-order explicit approximation and can become unstable.

8. **Record diagnostics.** Total represented mass and momentum, fluid divergence, drag work, collision energy change, concentrations, collision counts, and numerical convergence. Repeat with smaller timesteps, more parcels at fixed physical population, and changed widths and meshes.

This ordering is a first-order operator-splitting template. A second-order implementation needs a symmetric or otherwise validated higher-order coupling/splitting method. Periodic collision overlap is the sum over images; the free-space expression is valid only when image contributions are negligible or handled explicitly.

For fixed $\mathbf{U}_i$ and no other forces the exact drag substep is

$$\mathbf{v}_i^{n+1} = \mathbf{U}_i + (\mathbf{v}_i^n - \mathbf{U}_i)e^{-\Delta t/\tau_i}.$$

Deposit the opposite integrated impulse $-M_i(\mathbf{v}_i^{n+1} - \mathbf{v}_i^n)$ on the fluid for that drag substep. If locations or weights change appreciably, distribute it with consistent temporal quadrature. This exact frozen-fluid update does not by itself solve strongly coupled fluid evolution.

Useful resolution constraints include

$$\Delta t \ll \min\left(T_{\text{flow}}, \frac{h_i}{|\mathbf{v}_i|}, \lambda_{ij}^{-1}\right)$$

where relevant, together with the fluid solver's convective/viscous stability limits. Explicit drag additionally resolves τ_i. Soft-sphere contact additionally resolves the shortest contact time. For an unclipped underdamped linear spring–dashpot contact,

$$t_c = \frac{\pi}{\sqrt{k_n/\mu_{ij} - (\gamma_n/2\mu_{ij})^2}}.$$

Use many contact substeps per t_c and verify restitution numerically for the actual clipped or nonlinear law. There is no universal timestep or Gaussian width that is correct for every flow.

20 Verification before scientific use

Test	Required behavior or comparison
Kernel normalization	$\sum_a D_{ai}\Delta V_a = 1$ for moving parcels, including periodic seams
Uniform fluid sampling	A constant field is interpolated exactly
Drag relaxation	Match the exponential solution for fixed fluid velocity
Closed-domain momentum	Fluid plus represented particle momentum stays constant with no external force
Drag energy	Baseline coupling gives nonpositive drag work before temporal discretization error
Binary impact	Momentum conservation and the restitution/energy formulas in Section 11
Gaussian overlap	Numerical quadrature matches the analytical free-space overlap
Statistical collision rate	Ensemble mean approaches $N_i N_j K_{ij} I_{ij} \Delta t$ under frozen assumptions
Parcel refinement	Change multiplicities and parcel count at fixed physical number and initial population; recover stable statistics
Mesh and width study	Separate mesh convergence at fixed physical filter from sensitivity to the chosen filter model

Test	Required behavior or comparison
Wall and close approach	Verify wall impact and lubrication behavior if these physics are included
Concentrated benchmark	Compare to appropriate experiments or a validated dense-flow model before using a dense extension

A statistical method needs ensemble uncertainty estimates, not agreement from one random seed. Changing parcel count while also changing represented total mass is not a clean parcel-convergence test. A smooth solution and good momentum balance establish neither correct drag calibration nor correct collision statistics.

21 Recommended terminology for a thesis or implementation request

Use the following wording until an exact reference has been confirmed:

> Formulate an Eulerian–Lagrangian model for incompressible particle-laden flow. Represent the dispersed phase by weighted Lagrangian parcels with explicit physical particle size, density, multiplicity, position, and velocity. Couple the phases through normalized Gaussian interpolation and equal-and-opposite force spreading. Specify the force convention, disturbed or undisturbed slip velocity, kernel width, boundary treatment, and parcel refinement strategy. Define particle collisions separately using either a validated weighted stochastic algorithm, a calibrated coarse-grained contact model, or an MP-PIC stress closure. Derive conservation properties and validate drag, collision statistics, and mesh, timestep, and parcel convergence before application.

For the specific smooth-cloud interpretation, add: "Distinguish spatial density overlap from solid contact; state whether Gaussian overlap determines a collision rate or an effective interaction potential." For the FCM interpretation, instead request Gaussian force multipoles, mobility calibration, stresslet constraints, lubrication, and contact corrections. These descriptions identify different physical models and should not be used interchangeably.

22 Sources and reading guide

The references below were checked against publisher pages, author or institutional repositories, or official documentation. Direct full texts were available for R1, R2, R5, and R7; the RWTH author presentation was available for R3. Access to some publisher full texts and the NETL report was restricted, so only the supported framework or metadata is attributed to those sources. The note's baseline equations, overlap derivation, restricted equal-weight collision construction, conservation derivations, and Gaussian potential are explicitly specified here rather than claimed as exact reproductions of any one source.

R1. Apte, S. V., Mahesh, K., and Lundgren, T. (2008). Accounting for finite-size effects in simulations of disperse particle-laden flows. *International Journal of Multiphase Flow*, 34, 260–271. DOI: 10.1016/j.ijmultiphaseflow.2007.10.005. Author PDF. Closest overall match to Gaussian parcel-to-grid transfer plus separate collision and volume-displacement physics. Its particular coarse parcel contacts should not be assumed universally valid.

R2. Capecelatro, J., and Desjardins, O. (2013). An Euler–Lagrange strategy for simulating particle-laden flows. *Journal of Computational Physics*, 238, 1–31. DOI: 10.1016/j.jcp.2012.12.015. Author PDF. Read for volume-filtered conservation laws, closures, coupling consistency, and soft-sphere collisions.

R3. Pischke, P., Cordes, D., and Kneer, R. (2012). A collision algorithm for anisotropic disperse flows based on ellipsoidal parcel representations. *International Journal of Multiphase Flow*, 38, 1–16. DOI: 10.1016/j.ijmultiphaseflow.2011.09.002. See also Pischke and Kneer (2014), *A high resolution collision algorithm for anisotropic particle populations*, author presentation. Most targeted match if Gaussian influence regions were part of the collision-probability calculation. The full algorithm includes details beyond this note's simplified construction.

R4. Maxey, M. R., and Patel, B. K. (2001). Localized force representations for particles sedimenting in Stokes flow. *International Journal of Multiphase Flow*, 27, 1603–1626. Publisher record. Foundational force-envelope reference for FCM.

R5. Keaveny, E. E. (2014). Fluctuating force-coupling method for simulations of colloidal suspensions. *Journal of Computational Physics*, 269, 61–79. Author preprint. Read Section 3 for deterministic Gaussian envelopes, widths, velocity averages, and stresslets; later sections address thermal fluctuations.

R6. Dance, S. L., and Maxey, M. R. (2003). Incorporation of lubrication effects into the force-coupling method for particulate two-phase flow. *Journal of Computational Physics*, 189, 212–238. Institutional record. Shows why Gaussian hydrodynamic envelopes need additional close-range lubrication treatment.

R7. Gualtieri, P., Picano, F., Sardina, G., and Casciola, C. M. (2015). Exact regularized point particle method for multiphase flows in the two-way coupling regime. *Journal of Fluid Mechanics*, 773, 520–561. Publisher. Author preprint. ERPP derives regularization from viscous disturbance diffusion and treats self-induced disturbance; it is more specific than arbitrary Gaussian force deposition.

R8. Horwitz, J. A. K., and Mani, A. (2016). Accurate calculation of Stokes drag for point-particle tracking in two-way coupled flows. *Journal of Computational Physics*, 318, 85–109. Author preprint. Relevant to the distinction between resolved disturbed velocity and undisturbed velocity for drag.

R9. Clarke, M. A., and Musser, J. (2020). The MFiX Particle-in-Cell Method MFiX-PIC Theory Guide. DOE/NETL-2020/2115. DOI: 10.2172/1630414. NETL description. Consult for a complete dense parcel implementation and its conventions.

R10. LAMMPS official documentation. pair_style granular. Technical source for explicit granular normal and tangential contact laws and damping conventions. Accessed 15 October 2026.

R11. Shima, S., Kusano, K., Kawano, A., Sugiyama, T., and Kawahara, S. (2009). The super-droplet method for the numerical simulation of clouds and precipitation: a particle-based and probabilistic microphysics model coupled with a non-hydrostatic model. *Quarterly Journal of the Royal Meteorological Society*, 135, 1307–1320. DOI: 10.1002/qj.441. Author preprint. Relevant to weighted stochastic collisions and coalescence; not a generic solid-contact force law.

R12. Yade official documentation. CFD–DEM coupled simulations with Yade and OpenFOAM. Documents hydrodynamic coupling and Gaussian distribution of particle quantities to fluid cells. Accessed 15 October 2026.

R13. Maxey, M. R., and Riley, J. J. (1983). Equation of motion for a small rigid sphere in a nonuniform flow. *Physics of Fluids*, 26, 883–889. DOI: 10.1063/1.864230. Foundational small-sphere hydrodynamic force framework; its asymptotic assumptions remain essential.

R14. O'Rourke, P. J., and Snider, D. M. (2012). Inclusion of collisional return-to-isotropy in the MP-PIC method. *Chemical Engineering Science*, 80, 39–54. DOI: 10.1016/j.ces.2012.05.047. Publisher record. Relevant when the remembered Gaussian is a velocity distribution used in parcel collision relaxation.

R15. OpenFOAM Foundation official source. SchillerNaumannDragForce source. Direct implementation of the Schiller–Naumann drag correlation for Lagrangian particles. Accessed 15 October 2026.